BIO201 FINALTERM SUBJECTIVE MEGA FILE

ALL DATA FROM PAST PAPERS (2018-2022)

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FINAL TERM SUBJECTIVE (2022)

1- Regulator proteins and their types

Any protein that influences the regions of a DNA molecule that are transcribed by RNA polymerase during the process of transcription. These proteins, which include transcription factors, therefore help control the synthesis of proteins in cells.

Types:

The major regulatory thin-filament-associated proteins are tropomyosin (TM) and caldesmon (CaD), which associate with 14 actin monomers in ratios 2:1 (13).

2- 5 sub divisions of IgG

Four human IgG subclasses (IgG1, IgG2, IgG3, and IgG4), having γ1, γ2, γ3, and γ4 heavy chains, respectively.

3- Oncogenes

Upon activation, a proto-oncogene (or its product) Becomes a tumour inducing agent, an oncogene. Some oncogenes, usually involved in early stages of cancer development, increase the chance that a normal cell develops into a tumour cell, possibly resulting in cancer.

4- Endocytosis

The material to be ingested is progressively enclosed by a small portion of the plasma membrane, which first invaginates and then pinches off to form an endocytic vesicle containing the ingested substance or particle is called endocytosis.

Types:

- **Phagocytosis** involves the ingestion of large particles microorganisms or dead cells via large Vesicles called phagosomes (>250 nm in diameter)
- **Pinocytosis** involves ingestion of fluid & solutes via small pinocytic vesicles (about 100 nm in diameter).

5- Cell culture and its types

The growth of microorganisms such as bacteria and yeast, or human, plant, or animal cells in the laboratory. Cell cultures may be used to diagnose infections, to test new drugs, and in research.

Cells cultured in the lab can be classified into three different types:

- primary cells
- transformed cells
- self-renewing cells.

6- Proto oncogene.

A proto-oncogene is a normal gene that can become an oncogene, either after mutation or increased expression. Proto-oncogenes code for proteins that help to regulate cell growth and differentiation. Proto-oncogenes are often involved in signal transduction and execution of mitogenic signals, usually through their protein products. Some oncogenes, usually involved in early stages of cancer development, increase the chance that a normal cell develops into a tumour cell, possibly resulting in cancer

Classes of Oncogenes

- Growth Promoting Oncogenes
- Growth-Inhibiting Cancer Suppressor Genes
- DNA Repairing Genes
- Apoptosis Genes

7- Kin Recognition

ER resident proteins bind to one another, thus forming complexes that are too big to enter transport vesicles Aggregation of proteins that function in the same compartment—called kin recognition—is a general mechanism that compartments use to organize and retain their resident proteins.

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8- Difference btw T cells and B cell

T cells and B cells are both generated in the bone marrow from stem cells or more precisely form the lymphoid common progenitor. T cells or thymocytes maturate in the thymus, a lymphoid organ situated in the chest, while B cells mature in the bone marrow, at the same site of their generation.

Both T cells and B cells migrate to the bloodstream after their maturation and circulate between the peripheral lymphoid organs in an inactive state.

There are two types of activated T cells:

- Cytotoxic T cells responsible of the destruction of cells infected by intracellular pathogens
- helper T cells responsible of activating cytotoxic T cells, macrophages, and B cells.

B cells, on the other side, differentiate upon activation by helper T cells into one cell type, plasma cells capable of secreting antigen-specific antibodies.

Activated T cells present antigen receptors on their membrane and are not capable of secreting antibodies, whereas activated B cells are responsible for antibody secretion.

9- Signalling process

Signal processing involves converting or transforming data in a way that allows us to see things in it that are not possible via direct observation.

OR

The entire signalling process, from signal detection to final response is called signal transduction pathway/ signalling cascade.

10- Phagocytosis:

Phagocytosis involves the ingestion of large particles microorganism or dead cells via large vesicles called phagosomes (>250 nm in diameter).

11- Define Metaphase.

All the centromeres arrive at the equatorial plate. Metaphase is the best time to see the sizes and shapes of chromosomes because they are maximally condensed.

12- Function of ERGIC53

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The ERGIC53 protein serves as a receptor for packaging some secretory proteins into COPII coated vesicles. Its role in protein transport was identified because humans who lack it owing to an inherited mutation have lowered serum levels of two secreted blood-clotting factors (Factor V and Factor VIII) and therefore bleed excessively.

13- Role and function of intracellular receptors:

Intracellular receptors are macromolecules that may exist as free and soluble components of the cytoplasm, or may be contained within various intracellular compartments. These receptors function via their capacity to recognize and bind specific ligands. Once stimulated by lipid-soluble ligands, intracellular receptors translocate to the nucleus and bind to response elements of the spacer DNA, altering the rate of gene expression.

14- How UV Light Sterilized the Equipment.

UV sterilization also known as UV disinfection or ultraviolet germicidal irradiation (UVGI) works by breaking down certain chemical bonds and scrambling the structure of DNA, RNA and proteins, causing a microorganism to be unable to multiply. Ultraviolet (UV) radiation, under certain conditions, serves as an excellent sterilant. Bacteria, yeasts, molds and viruses are readily inactivated, provided the cells are accessible to a sufficient UV dose. The biocidal effectiveness of UV radiation emitted from three different lamps was determined.

15- Differentiate Between MCH-I and MCH-II.

There are two main structurally & functionally distinct classes of MHC proteins:

- **Class I:** class I MHC proteins, present foreign peptides to cytotoxic T cells, & class II present foreign peptides to helper T cells. 3 genes for class I proteins (HLA-A, B &C).
- **Class II:** 3 genes for class II proteins (HLA-DP, DQ &DR) genes each for both the classes. Class II can α & β maternal & paternal can pair up Polymorphism VS Gene Rearrangement.

A person makes a small number of MHC proteins MHC should be able to present peptide fragments from almost any foreign protein to T cells MHC groove can accommodate an extended peptide about 8–10 amino acids long. In extended conformation terminal amino group binds to an invariant pocket at one end of the groove and its terminal carboxyl group bound to an invariant pocket at the other end of the groove. Pockets recognize peptide backbone features common to all peptides, each class I MHC protein can bind a peptides of diverse sequence.

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16. Explain the process of Endocytosis.

The material to be ingested is progressively enclosed by a small portion of the plasma membrane, which first invaginates and then pinches off to form an endocytic vesicle containing the ingested substance or particle.

17. Operons

Cluster of genes with 1 promoter is called operon. Operon consists of a promoter, an operator, and two or more structural genes. The operator lies between the promoter & structural genes binds regulatory proteins that either activate or repress transcription.

18. Structure of antibodies:

It is composed of four polypeptide chains:

- 2 identical heavy chains
- 2 identical light chains.

Tail (Fc) & hinge region are formed by the two heavy chains. 2 identical Binding Sites. Flexible hinge region improves efficiency of Antigen binding and cross-linking. Immunoglobin (Ig) domain is 110 AA long and is repeated suggesting gene duplication.

19. Helper T cells

Effector helper T cells stimulate the responses of other cells phagocytic macrophages, B cells, and cytotoxic T cells.

20. Inclusion cell disease

In this disease almost all of the hydrolytic enzymes are missing from the lysosomes of fibroblasts, and their undigested substrate accumulate in lysosomes, which consequently form large “inclusions” in the cells. In I-cell disease patients, all the hydrolases missing from Lysosomes are found in the blood. Because they fail to be sorted properly in the Golgi apparatus, the hydrolases are secreted rather than transported to lysosomes.

21. Proteins structure

I. Primary Structure: -

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Sequence of amino acids bonded by peptide linkages (Diversity 20n).

II. Secondary Structure

- α helices (hydrogen bonds between amino acid residues).
- β pleated sheets (hydrogen bonds between amino acid residues).

III. Tertiary Structure

Generated by bending and folding of the polypeptide chain.

IV. Quaternary structure

Arrangement of multiple folded protein molecules in a multisubunit complex.

22. Why cell signalling is important

In order to survive even simplest organisms, need to sense and respond to their environment. It is critical that the cells of multicellular organisms communicate to coordinate their efforts (Running). Multicellular organism cells are specialized and rely on each other for the support (brain sugar). During development there have to be checks balances on differentiation (analogy society). Our sense organs allow us to respond to light, odors and tastes (chemical), temperature, touch, and sound. Magnetism can be a signal: some bacteria and birds orient themselves to Earth's magnetic poles.

23. Olfaction

Olfaction is a critical physiologic process of the nasal airway. The sense of smell plays an important role in pleasure, including the palatability and flavour of food, and can even modify dietary behaviour and nutrition, a key component of health.

24. Monolayer conflent and past confluency

Cells should be sub-cultured just before they approach confluences. A confluent monolayer is an adherent cell culture (dish, plate or flask) in which the cells have formed a single layer over the entire surface area available for growth. Once the cells have started to form cluster above the first layer or have started to lift up from the surface, the culture is past confluency stage. Cells that are repeatedly passaged at densities past confluency display decreased doubling times, decreased viabilities, & decreased ability to attach.

25. Why cell is important

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Cells provide structure and function for all living things, from microorganisms to humans. Scientists consider them the smallest form of life. Cells house the biological machinery that makes the proteins, chemicals, and signals responsible for everything that happens inside our bodies.

26. Name the antibody that appear in the surface of the B-cell?

IgM first class of antibody to appear on the surface of a developing B cell. The major class secreted in the early stages of a primary antibody response. Secreted form, IgM is a pentamer, J chain is required for pentamer formation. The binding of an antigen to a single secreted pentameric IgM molecule can activate the complement system, which can either mark the pathogen for phagocytosis or kill it directly.

27. Transduction proteins

Changes in protein level, protein localization, protein activity, and protein–protein interactions are critical aspects of signal transduction, allowing cells to respond highly specifically to a nearly limitless set of cues and also to vary the sensitivity, duration, and dynamics of the response.

SUBJECTIVE FROM PAST PAPERS

1. ER single sequence is guided to the ER by which of the two components?

- signal recognition particles
- srp receptor

2. What is the role of regulatory region explaining its two elements?

- Regulatory sequence controls when expression occurs for the multiple protein coding regions (red).
- Promoter, operator and enhancer regions (yellow) regulate the transcription of the gene.

3. Describe the two signalling functions of N-terminal ER signal sequence of a soluble protein?

The N-terminal ER signal sequence of a soluble protein has two signalling functions:

1) It directs the protein to the ER membrane

2) It serves as a start-transfer signal (or start-transfer peptide) that opens the pore.

4. Cell Cycle (10 marks)

The cell cycle or cell-division cycle is the series of events that take place in a cell leading to its division and duplication of its DNA (DNA replication) to produce two daughter cells. In bacteria, which lack a cell nucleus, the cell cycle is divided into the B, C, and D periods. The B

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period extends from the end of cell division to the beginning of DNA replication. DNA replication occurs during the C period. The D period refers to the stage between the end of DNA replication and the splitting of the bacterial cell into two daughter cells.

In cells with a nucleus, as in eukaryotes, the cell cycle is also divided into three periods:

- Interphase
- the mitotic (M) phase
- cytokinesis.

During interphase, the cell grows, accumulating nutrients needed for mitosis, preparing it for cell division and duplicating its DNA.

During the mitotic phase, the chromosomes separate.

During the final stage, cytokinesis, the chromosomes and cytoplasm separate into two new daughter cells. To ensure the proper division of the cell, there are control mechanisms known as cell cycle checkpoints.

The cell-division cycle is a vital process by which a single-celled fertilized egg develops into a mature organism, as well as the process by which hair, skin, blood cells, and some internal organs are renewed. After cell division, each of the daughter cells begin the interphase of a new cycle. Although the various stages of interphase are not usually morphologically distinguishable, each phase of the cell cycle has a distinct set of specialized biochemical processes that prepare the cell for initiation of cell divisions.

- prophase
- metaphase
- anaphase
- telophase
-

5. antibodies names?

- lgA antibody
- lgD antibody
- lgE antibody
- lgG antibody
- lgM antibody

6. what is stem cell and its benefits and therapeutic uses?

Stem Cell: -

An undifferentiated cell of a multicellular organism which is capable of giving rise to indefinitely more cells of the same type, and from which certain other kinds of cell arise by differentiation.

Benefits: -

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The hope going forward is that stem cells can also be used as a "renewable source of replacement cells and tissues" to treat common and serious diseases without the need for organ transplants or surgeries, including: macular degeneration, spinal cord injury, stroke, burns, heart disease, diabetes, osteoarthritis.

Uses: -

Stem-cell therapy is the use of stem cells to treat or prevent a disease or condition. Bone marrow transplant is the most widely used stem-cell therapy, but some therapies derived from umbilical cord blood are also in use.

7. Steps of RNA processing?

The eukaryotic pre-mRNA undergoes extensive processing before it is ready to be translated.

- 5' Capping.
- 3' Poly-A Tail.
- Pre-mRNA Splicing.
- Discovery of Introns.
- Intron Processing.

8. Mechanism of the olfaction receptors?

It can detect thousands of different odour molecules through a large family of olfactory receptors (ORs), of a diverse protein sequence, which are located in olfactory sensory neurons (OSNs) in the olfactory epithelium in the nose of humans, and in the vomer nasal organ in animals. Odorant molecule pass through the nasal cavity.

1: binding of odorant to its receptor activates G protein

2: the activation of G protein activates the synthesis of aAMP by adenylyl cyclase.

3: cAMP activates the opening site of ion channel

4: change in the ion concentration inside the cell send a signal to a specific area of the brain perceives the signal as ascent.

9. How G protein involved in formation of GDP?

Rhodopsin absorbs the light causing the G protein transduction to exchange GTP for GDP.

10. How sheep is cloned by the nuclear material. Explain it by all steps. 5 marks

Nuclear transfer is a form of cloning. The steps involve removing the DNA from an oocyte (unfertilised egg), and injecting the nucleus which contains the DNA to be cloned. In rare instances, the newly constructed cell will divide normally, replicating the new DNA while remaining in a pluripotent state. If the cloned cells are placed in the uterus of a female mammal, a cloned organism develops to term in rare instances. This is how Dolly the Sheep and many other species were cloned. Cows are commonly cloned to select those that have the best milk

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production. Despite this, the low efficiency of the technique has prompted some researchers, notably Ian Wilmot, creator of Dolly the cloned sheep, to abandon it.

11. Write synapse and its steps involved in the mechanism. 5 marks

A junction between two nerve cells, consisting of a minute gap across which impulses pass by diffusion of a neurotransmitter.

Steps: -

There are two general classes of neurotransmitters: large neuropeptides or smaller amines/amino acids which are synthesized differently.

Step 1 – Neurotransmitter Synthesis.

Step 2 & 3 - Neurotransmitter Packaging and Release.

Step 4 – Neurotransmitter Binding.

Step 5 – Stopping the Chemical Signal.

Mechanism: -

Neurotransmitters, such as Acetylcholine, are stored in vesicles in the presynaptic terminal. Acetylcholine is secreted across synaptic clefts, where it binds to the post synaptic membrane and causes a change in the permeability of the postsynaptic cell, which results in an excitatory post-synaptic potential.

12. TGF-ß Receptor Family. 3 marks

Transforming growth factor beta (TGFβ) receptors are single pass serine/threonine kinase receptors. They exist in several different isoforms that can be homo- or heterodimeric. The number of characterized ligands in the TGFβ superfamily far exceeds the number of known receptors, suggesting the promiscuity that exists between the ligand and receptor interactions.

13. CAm kinase function

CaM kinaseII or CaMKII is a serine/threonine-specific proteinkinase that is regulated by the Ca 2+ /calmodulin complex. CaMKII is involved in many signaling cascades and is thought to be an important mediator of learning and memory.

14. Homotypic and heterotypic fusion

Heterotypic fusion occurs between compartments that differ in composition. For example, a vesicle derived from one compartment fuses heterotypically with a different compartment.

Homotypic fusion occurs between compartments that are biochemically identical. For example, endosomes fuse with one another, as do lysosomes, the ER and the mitochondria.

15. Confluent monolayer & past confluent

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Confluent monolayer refers to cells in tissue culture, e.g. epithelial cells, which form a cohesive sheet comprising of a single cell layer filling the entire surface area of the bottom of the culture dish. Cells have then reached confluence.

16. Banzer exp result

The T4 rII system is an experimental system developed in the 1950s by Seymour Benzer for studying the substructure of the gene. The experimental system is based on genetic crosses of different mutant strains of bacteriophage T4, a virus that infects the bacteria E. coli.

17. 5 types of antibiotic

IgM, IgD, IgG, IgE, IgA

18. Restriction endonuclease

An enzyme produced chiefly by certain bacteria that has the property of cleaving DNA molecules at or near a specific sequence of bases.

19. Pcr requirements

The various components required for PCR include a DNA sample, DNA primers, free nucleotides called ddNTPs, and DNA polymerase. The various components required for PCR include a DNA sample, DNA primers, free nucleotides called ddNTPs, and DNA polymerase.

20. Sheep cloned through nuclear transfer

"Sheep Cloned by Nuclear Transfer from a Cultured Cell Line" (1996), by Keith Campbell, Jim McWhir, William Ritchie, and Ian Wilmut. ... This experiment led the Roslin team to later clone mammals from adult body cells and to genetically engineer mammals

21. 1Antibodies on surface of B cells

IgM, IgD

22. Peptide bond

A peptide bond is an amide type of covalent chemical bond linking two consecutive alphaamino acids from C1 (carbon number one) of one alpha-amino acid and N2 (nitrogen number two) of another along apeptide or protein chain.

23. T cells B cells difference

An important difference between T-cells and B-cellsis that B-cells can connect to intact antigens right on the surface of the invading virus or bacteria. This isdifferent from Tcells, which can only connect to parts of virus antigens on the outside of infected cells.

24. Function of centromere in cell division

Centromere Functions. One major function of acentromere is joining the sister chromatids. ... On each chromatid, the kinetochore forms at the centromere region of the DNA. Once all of the

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chromatids are attached to the mitotic spindle, the microtubules pull the sister chromatids apart into the two future daughter cells

25. Missense mutation

In genetics, a missense mutation is a point mutationin which a single nucleotide change results in a codon that codes for a different amino acid. It is a type of nonsynonymous substitution.

26. Function of n terminal singnaling of ER

The N-terminus is the start of a protein or polypeptide referring to the free amine group (-NH2) ... 1 Chemistry; 2 Function ... The N-terminal signal peptide is recognized by the signal recognition particle (SRP) and results in the targeting ... In eukaryotic cells, these proteins are synthesized at the rough endoplasmic reticulum.

27. 19.Gel electrophoresis

Gel electrophoresis is a method for separation and analysis of macromolecules and their fragments, based on their size and charge.

28. What is meant by polarity of water??

Water is a "polar" molecule, meaning that there is an uneven distribution of electron density. Water has a partial negative charge near the oxygen atom due the unshared pairs of electrons, and partial positive charges near the hydrogen atoms.

29. Examples of RNA viruses

This nucleic acid is usually single-stranded RNA (ssRNA) but may be double-stranded RNA (dsRNA).Notable human diseases caused by RNA viruses include Ebola hemorrhoragic fever, SARS, Rabies,common cold, influenza, hepatitis C, West Nile fever, polio and measles.

30. Paracrine signaling

Paracrine signaling is a form of cell-to-cell communication in which a cell produces a signal to induce changes in nearby cells, altering the behavior of those cells.

31. Down syndrome

a congenital disorder arising from a chromosome defect, causing intellectual impairment and physical abnormalities including short stature and a broad facial profile. It arises from a defect involving chromosome 21, usually an extra copy (trisomy-21).

32. Regulatory regions

Regulatory sequence. A regulatory sequence is a segment of a nucleic acid molecule which is capable of increasing or decreasing the expression of specific genes within an organism. Regulation of gene expression is an essential feature of all living organisms and viruses.

33. ER signal guid ER by which of two components?

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The ER signal sequence is guided to the ER membrane by at least two components. SignaI-recognition particle (SRP): cycles between the ER membrane and the cytosol and binds to the signal sequenceo SRP receptor. The SRP consists of six different polypeptide chains to a single small RNA molecule. Each ER signal sequence has eight or more nonpolar amino acids at its center.

34. Polypeptide and anupolied?

Polypeptide a linear organic polymer consisting of a large number of amino-acid residues bonded together in a chain, forming part of (or the whole of) a protein molecule. Aneuploidy is the presence of an abnormal numbers of chromosomes in a cell, for example a human cell having 45 or 47 chromosomes instead of the usual 46.

35. Adult stem cells and theirs beneficial and therapeutic uses ?

Adult stem cells have been identified in many organs and tissues, including brain, bone marrow, peripheral blood, blood vessels, skeletal muscle, skin, teeth, heart, gut, liver, ovarian epithelium, and testis. They are thought to reside in a specific area of each tissue (called a "stem cell niche"). Stem cell approaches have shown great promise for regeneration in models of glaucoma. Multipotent stem cells are retrieved from tissues such as bone marrow and fat, thus avoiding ethical concerns.

Vision loss from glaucoma occurs when axons in the optic nerve become damaged and can no longer carry visual information to the brain. However, these treatments can only preserve remaining vision; they don't improve or restore vision that already has been lost due to glaucoma

36. How nuclear transformations occur in the sheep process..?

Nuclear transfer is a form of cloning. The steps involve removing the DNA from an oocyte (unfertilised egg), and injecting the nucleus which contains the DNA to be cloned. In rare instances, the newly constructed cell will divide normally, replicating the new DNA while remaining in a pluripotent state.

37. Elaborate the mechanism of growth promoting mechanism of oncogenes. 10 marks

The activation of oncogenes involves genetic changes to cellular proto-oncogenes. The consequence of these genetic alterations is to confer a growth advantage to the cell.

Three genetic mechanisms activate oncogenes in human neoplasms:

(1) Mutation

(2) Gene amplification,

(3) Chromosome rearrangements.

These mechanisms result in either an alteration of proto-oncogene structure or an increase in proto-oncogene expression because neoplasia is a multistep process, more than one of these

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mechanisms often contributes to the genesis of human tumours by altering a number of cancer-associated genes. Full expression of the neoplastic phenotype, including the capacity for metastasis, usually involves a combination of proto-oncogene activation and tumour suppressor gene loss or inactivation.

38. The Endoplasmic Reticulum is guided to ER by which 2 components.

At least two cytoplasmic components are involved in the specific targeting of nascent secretory and membrane proteins to the ER, the signal recognition particle (SRP) and the docking protein (DP) or SRP receptor.

39. TGF-& Receptor Family. 3 marks

The Transforming Growth Factor beta (TGF) receptors are a family of serine/threonine kinase receptors involved in TGF beta signalling pathway. These receptors bind growth factor and cytokine signalling proteins such as TGF§s (TGFB1 , TGFB2, TGFB3), bone morphogenetic proteins (BMPs), growth differentiation factors (GDFs), activin and inhibin, myostatin, anti-Mullerian hormone (AMH), and NODAL, Type II, Type III.

40. Inhibitors that binds to the active side of enzymes are

The competitive inhibitor resembles the substrate and binds to the active site of the enzyme the substrate is thereby prevented from binding to the same active site. A competitive inhibitor diminishes the rate of catalysis by reducing the proportion of enzyme molecules bound to a substrate.

41. Inhibitors that bind to the non-active side of enzymes are

A non-competitive inhibitor binds to a different site that is not the active site of the enzyme and changes the structure of the enzyme; therefore, it blocks the enzyme from binding to substrate, which stops enzyme activity.

42. Phosphate and non phosphate group?

A phosphate group is just a phosphorus atom bound to four oxygen atoms, but it has many important roles. Along with sugars and bases, it makes up nucleic acids, like DNA and RNA. As part of energy carriers, like ATP, it provides energy for moving our muscles.

43. Antibody that appear on the surface of developing B cell?.what is the role of regulatory region explain its two elements? (REPEAT)

IgM is not only the first class of antibody to appear on the surface of a developing B cell. It is also the major class secreted into the blood in the early stages of a primary antibody response, on first exposure to an antigen.

The specific variable (V)-region and constant (C)-region coding sequences that the tumor cells used were present on the same DNA restriction fragment in the tumor cells but on two different restriction fragments in the embryos. This showed that the DNA sequences encoding an antibody molecule are rearranged at some stage in B cell development

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44. Describe the two signaling functions of N-terminal ER signal sequence of a soluble protein?

A sequence of hydrophobic amino acids, the ER signal sequence, is recognized and bound by a signal-recognition particle (SRP), which in turn is bound by an SRP receptor on the rough ER membrane. Generally, ER signal sequences are located at the N- terminus and are cleaved from the protein in the rough ER lumen.

45. Functions of growth promotor oncogenes?

An oncogene is a gene that has the potential to cause cancer.'1' In tumor cells, they are often mutated and/or expressed at high levels.

Most oncogenes began as proto-oncogenes, normal genes involved in cell growth and proliferation or inhibition of apoptosis. If normal genes promoting cellular growth, through mutation, are up-regulated, (gain of function mutation) they will predispose the cell to cancer and are thus termed oncogenes.

46. Process of translation?

In molecular biology and genetics, translation is the process in which ribosomes in the cytoplasm or ER synthesize proteins after the process of transcription of DNA to RNA in the cell's nucleus. The entire process is called gene expression.

47. Which mechanism is used in certain birds destination?

Birds use the Earth's magnetic field to navigate but scientists are still unsure of how it works. About 50 animal species, ranging from birds and mammals to reptiles and insects, use Earth's magnetic field for navigation.

48. Enlist the function of signaling system ?

Cell signalling in multicellular organisms. In a multicellular organism, signalling between cells occurs either through release into the extracellular space, divided in paracrine signalling (over short distances) and endocrine signalling (over long distances), or by direct contact, known as juxtacrine signaling.

49. Describe the principle and air flow chart?

Flow of air or any other fluid is caused by a pressure differential between two points. Flow will originate from an area of high energy, or pressure, and proceed to area(s) of lower energy or pressure.

50. Examples of 3 RNA viruses?

Hepatitis C virus, measles virus, rabies virus, influenza virus and Ebola virus are examples of RNA virus.

51. Name 5 major growth promoting oncogenes? Epidermal growth factor (EGF)

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Transforming growth factor alpha (TGFa) epidermal growth factor receptor (EGFR) c-erb B-2, Cytoplasmatic mediators K-ras, H-ras, src

52. Translation process?

In molecular biology and genetics, translation is the process in which ribosomes in the cytoplasm or ER synthesize proteins after the process of transcription of DNA to RNA in the cell's nucleus. ... In translation, messenger RNA (mRNA) is decoded in a ribosome to produce a specific amino acid chain, or polypeptide.

53. Endocytosis and phagocytosis or pinocytosis?

Endocytosis and its counterpart, exocytosis, are used by all cells because most chemical substances important to them are large polar molecules that cannot pass through the hydrophobic plasma or cell membrane by passive means. Endocytosis includes pinocytosis (cell drinking) and phagocytosis (cell eating).

54. Laminar cabinet flow principle?

A laminar flow cabinet or laminar flow closet or tissue culture hood is a carefully enclosed bench designed to prevent contamination of semiconductor wafers, biological samples, or any particle sensitive materials. Air is drawn through a HEPA filter and blown in a very smooth, laminar flow towards the user.

55. Describe process of translation briefly?

Translation is the process of translating the sequence of a messenger RNA (mRNA) molecule to a sequence of amino acids during protein synthesis. The genetic code describes the relationship between the sequence of base pairs in a gene and the corresponding amino acid sequence that it encodes.

56. Describe role of MHC in immune system. Differentiation be it's 2 types?

The major histocompatibility complex (MHC) is a set of cell surface proteins essential for the acquired immune system to recognize foreign molecules in vertebrates, which in turn determines histocompatibility. ... The human MHC is also called the HLA (human leukocyte antigen) complex (often just the HLA).

Professional antigen presenting cells (APCs) are immune cells that specialize in presenting an antigen to a T-cell. ... A professional APC takes up an antigen, processes it, and returns part of it to its surface, along with a class II major histocompatibility complex (MHC).

57. What are Stress elements?

Stress elements are a useful way to represent stresses acting at some point on a body. Isolate a small element and show stresses acting on all faces. Dimensions are “infinitesimal”, but are drawn to a large scale.

58. How &-blocker effect the function of adrenaline?

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Beta blockers work by blocking the effects of the hormone epinephrine, also known as adrenaline. When you take beta blockers, your heart beats more slowly and with less force, thereby reducing blood pressure. Beta blockers also help blood vessels open up to improve blood flow.

59. What is function of Rod cells?

Rod cells are photoreceptor cells in the retina of the eye that can function in less intense light than the other type of visual photoreceptor, cone cells. Rods are usually found concentrated at the outer edges of the retina and are used in peripheral vision.

60. Short note on transcription?

Transcription is the first step of gene expression, in which a particular segment of DNA is copied into RNA by the enzyme RNA polymerase. Both DNA and RNA are nucleic acids, which use base pairs of nucleotides as a complementary language.

61. Elaborate the point increased CAMP activity?

Cyclic adenosine monophosphate (cAMP, cyclic AMP, or 3',5'-cyclic adenosine monophosphate) is a second messenger important in many biological processes. cAMP is a derivative of adenosine triphosphate (ATP) and used for intracellular signal transduction in many different organisms, conveying the cAMP-dependent pathway.

62. What is RFLPS? Write its function?

In molecular biology, restriction fragment length polymorphism is a technique that exploits variations in homologous DNA sequences RFLP analysis was an important tool in genome mapping, localization of genes for genetic disorders, determination of risk for disease, and paternity testing.

63. CaM kinase 2 involve in muscles movement explain?

Calmodulin and its roles in skeletal muscle function. ... The overall result of this protein- protein interaction is a physiological effect, e.g., Ca2+ binding to calmodulinin smooth muscle allow it to interact with and activate myosin light chain kinase which catalyses the phosphorylation of myosin.

64. Differentiate the confluent monolayer and past confluent stages?

Confluent monolayer refers to cells in tissue culture, e.g. epithelial cells, which form a cohesive sheet comprising of a single cell layer filling the entire surface area of the bottom of the culture dish. Cells have then reached confluence.

65. Briefly describe the signal involved in olfaction?

Olfaction is a chemoreception that forms the sense of smell. Olfaction has many purposes, such as the detection of hazards, pheromones, and food. It integrates with other senses to form the sense of flavor. Olfaction occurs when odorants bind to specific sites on olfactory receptors

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located in the nasal cavity.“ Glomeruli aggregate signals from these receptors and transmit them to the olfactory bulb, where the sensory input will start to interact with parts of the brain responsible for smell identification, memory, and emotion.

66. Adapter hypothesis and messenger hypothesis?

The adaptor hypothesis is part of a scheme to explain how information encoded in DNA is used to specify the amino acid sequence of proteins. It was formulated by Francis Crick in the mid-1950s, together with the central dogma of molecular biology and the sequence hypothesis.

One of the seminal advances in this field came in the late 1950s when Earl Sutherland and his colleagues demonstrated that epinephrine induces glycogenolysis in the liver by stimulating the synthesis of an intracellular second messenger, 3’,5’-cyclic adenosine monophosphate (cAMP).

67. What is the role of regulatory regions explain its two elements?

A regulatory sequence is a segment of a nucleic acid molecule which is capable of increasing or decreasing the expression of specific genes within an organism. Regulation of gene expression is an essential feature of all living organisms and viruses.

68. Housekeeping gene Differential genes?

In molecular biology, housekeeping genes are typically constitutive genes that are required for the maintenance of basic cellular function, and are expressed in all cells of an organism under normal and patho-physiological conditions

Based on the embryological evidence for genomic equivalence (and on bacterial models of gene regulation), a consensus emerged in the 1960s that cells differentiate through differential gene expression.

69. Conditional genes?

Conditional gene knockout is a technique used to eliminate a specific gene in a certain tissue, such as the liver. This technique is useful to study the role of individual genes in living organisms.

70. Euchromatin and heterochromatin?

The DNA in the nucleus exists in two forms that reflect the level of activity of the cell. Euchromatin is prevalent in cells that are active in the transcription of many of their genes while heterochromatin is most abundant in cells that are less active or not active.

71. Write short note on COPI?

COPI is a co-atomer, a protein complex that coats vesicles transporting proteins from the cis end of the Golgi complex back to the rough endoplasmic reticulum, where they were originally synthesized, and between Golgi compartments.

72. Function of homotypic acid oxidase?

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Nitric oxide provokes vasodilation and inhibits platelet aggregation. We examined the effect of nitric oxide on superoxide anion production by three sources: activated intact neutrophils, xanthine oxidase/hypoxanthine, and the NADPH oxidase

73. Homotypic and heterotypic fusion and how they form Exocytosis?

There are two different types of cell fusion that can occur. These two types include homotypic and heterotypic cell fusion. Homotypic cell fusion occurs between cells of the same type. ... A heterokaryon is the melding of two or more cells into one and it may reproduce itself for several generations.

In exocytosis, membrane-bound secretory vesicles are carried to the cell membrane, and their contents (i.e., water-soluble molecules) are secreted into the extracellular environment. This secretion is possible because the vesicle transiently fuses with the outer cell membrane.

74. Mechanism of homogenetic oxydase?

Homogeneous catalytic O2 reduction to water by a cytochrome c oxidase model is done by trapping of intermediates and mechanistic insight.

75. Benzer experiment?

The T4 rll system is an experimental system developed in the 1950s by Seymour Benzer for studying the substructure of the gene. The experimental system is based on genetic crosses of different mutant strains of bacteriophage T4, a virus that infects the bacteria E. coli.

76. Signaling factors?

Many growth factors bind to receptors at the cell surface and stimulate cells to progress through the cell cycle and divide. Many cellular proteins are activated downstream of the growth factor receptors (such as EGFR) that initiate this signal transduction pathway.

77. What is E.coil cells?

E. coli is a Gram-negative rod-shaped bacteria, which possesses adhesive fimbriae and a cell wall that consists of an outer membrane containing lipopolysaccharides, a periplasmic space with a peptidoglycan layer, and an inner, cytoplasmic membrane.

78. Define somatic mutation?

An alteration in the DNA that occurs after conception. Somatic mutations can occur in any of the cells of the body except the germ cells (sperm and egg) and therefore are not passed on to children. These alterations can (but do not always) cause cancer or other diseases.

79. Write 3 names of stop codon?

In the standard genetic code, there are three different stop codons:

in RNA: UAG ("amber") UAA ("ochre") UGA ("opal")

in DNA: TAG ("amber") TAA ("ochre") TGA ("opal" or "umber")

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80. Deffirence between heterotypic nd homotypic?

Medical Definition of heterotypic:

1: of or being the reduction division of meiosis as contrasted with typical mitotic division — compare homotypic

2: 2 different in kind, arrangement, or form. heterotypic aggregations of cells.

81. Antibody used in I cell?

An anti-CD3 monoclonal antibody is one that binds to CD3 on the surface of T cells. They are immunosuppresive drugs. The first to be approved was muromonab-CD3in 1986, to treat transplant rejection. Newer monoclonal antibodies with the same mechanism of action include otelixizumab, teplizumab and visilizumab.

82. Use of regulatory system Explain 2 types?

A two-component regulatory system is a stimulus-response coupling mechanism that allows bacteria to sense and respond to changes in different environmental conditions

83. Role and function of intracellular receptors?

Hormones with Intracellular Receptors; Receptors for steroid and thyroid hormones are located inside target cells, in the cytoplasm or nucleus, and function as ligand- dependent transcription factors. ... Thus, the mechanism of action of steroid hormones is to modulate gene expression in target cells.

84. Andocytoces and phagosytosis df panocytoces?

Endocytosis and its counterpart, exocytosis, are used by all cells because most chemical substances important to them are large polar molecules that cannot pass through the hydrophobic plasma or cell membrane by active means. Endocytosis includes pinocytosis (cell drinking) and phagocytosis (cell eating).

85. Pinocytosis?

The ingestion of liquid into a cell by the budding of small vesicles from the cell membrane is called as pinocytosis.

86. UV light sterilization

UV light sterilization is an environmentally friendly method of killing bacteria, molds, fungi, and viruses without the use of harmful chemicals and does not produce corrosive materials or disinfection by-products (DBPs).

87. Enlist 5 types of antibiotics

These are the main types of antibiotics.

Penicillin:- such as penicillin and amoxicillin

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Cephalosporin:-such as cephalexin(Keflex)

Macrolides:- such as erythromycin (EMycin), clarithromycin(Biaxin), and azithromycin(Zithromax)

Fluoroquinolones:- such as ciprofloxacin (Cipro), levofloxacin (Levaquin), and ofloxacin (Floxin)

Sulfonamides:- such as co-trimoxazole(Bactrim) and trimethoprim(Proloprim)

Tetracyclines:- such as tetracycline (Sumycin,Panmycin) and doxycycline (Vibramycin)

Aminoglycosides:- such as gentamicin(Garamycin) and tobramycin(Tobrex)

88. Confluent monolayer

Confluent monolayer refers to cells in tissue culture e.g: epithelial cells which form a cohesive sheet comprising of a single cell layer filling the entire surface area of the bottom of the culture dish. Cell shave then reached confluence.

89. Write the name of antibody that appears on the surface of developing B cell?

IgM is the 1st class of Ab to appear on cell surface and secreted B cells can then switch Ab class and IgD.

90. Describe the two signalling functions of N-terminal ER signal sequence of a soluble protein

The N-terminal ER signal sequence of a soluble protein has two signalling functions 1) It directs the protein to the ER membrane 2) It serves as a start-transfer signal (or start-transfer peptide) that opens the pore.

91. Elaborate the mechanism of growth promoting mechanism of oncogenes? (REPEAT)

Growth Promoting Oncogenes:

a) Growth Factors b) Growth Factor Receptors c) Signal-Transducing Proteins (secondary messengers) d) Nuclear Transcription Factors e) Cyclins and Cyclin Dependent Kinases (CDKs) Autonomous Cell Growth In Cancer Cells Is Promoted By Following Elements

A) Growth Factor

Some cancer cells acquire the ability through mutations to produce growth factors to which they are responsive (autocrine signaling). Many glioblastomas secrete PDGF and sarcomas make TGF-a. Similar loops are fairly common many types of cancers.

B) Growth Factor Receptors

Mutated Growth Factor Receptors continue to deliver continues signals in the absence of the ligand. EGF receptor ERBB1 is over expressed in 80% of SCC of lung. ERBB2/HER2 is

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amplified in 25%-30% of breast cancers and adenocarcinomas of lung , ovary and salivary glands. These tumours are sensitive to small amounts of growth factors.

c) Signal-Transducing Proteins (Secondary Messengers)

Mutaion of RAS is the most common oncogene abnormality, approximately 30% of all human tumors contain mutated version of RAS gene. GAP/neurofibramin 1 (a RAS deactivating protein) is mutated in neurofibramatosis type 1. BCR-ABL hybrid gene activates RAS pathways (like p53 Normal ABL gene also promotes apoptosis via NLS domain).

D) Nuclear Transcription Factors

MYC and the polymerase assemble on a specific DNA sequence and drive transcription of genes. MYC protooncogene is expressed in virtually all the cells, & the MYC protein is rapidly induced when cells receive a signal to divide. MYC binds DNA and causes transcriptional activation of many growth related proteins. Dysregulation of of MYC gene occurs in Burkitt Lymphoma (B-cell tumor); MYC is also amplified in many other cancers

Mutated Growth Factor Receptors continue to deliver continues signals in the absence of the ligand. EGF receptor ERBB1 is over expressed in 80% of SCC of lung. ERBB2/HER2 is amplified in 25%-30% of breast cancers and adenocarcinomas of lung, ovary and salivary glands. These tumours are sensitive to small amounts of growth factors.

Mutaion of RAS is the most common oncogene abnormality, approximately 30% of all human tumors contain mutated version of RAS gene. GAP/neurofibramin 1 (a RAS deactivating protein) is mutated in neurofibramatosis type 1. BCR-ABL hybrid gene activates RAS pathways (like p53 Normal ABL gene also promotes apoptosis via NLS domain).

92. Describe TGF-B Receptor Family.

It consists of a large number of structurally related, secreted, dimeric proteins. During development, they regulate pattern formation and influence various cell behaviors, ie proliferation, differentiation, extracellular matrix production, and cell death. In adults, tissue repair and in immune regulation is the main funcation. Act through enzyme-linked receptors that are single-pass transmembrane proteins. Ligand first binds and activates a type-ll receptor homodimer, which recruits, phosphorylates, and activates a type-l receptor homodimer, forming an active tetrameric receptor complex Activated TGF-B receptors phosphorylate Smad2 activated BMP receptors phosphorylate Smadı, Smad5, or Smad8.One of these Smads then dissociates from the receptor and binds to Smad 4. Both type-l and typell receptors, required for signaling. The type-l receptor directly binds and phosphorylates a latent gene regulatory protein of Smad family

93. Define Electronegativity ?.

The tendency of an atom to attract electrons towards itself is called electronegativity.

94. What is Chargaff rule?

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In DNA, the amount of Purines (A+G) is always equal to the amount of pyrimidine (T+C)

95. Write mitochondrial features?

Mitochondria are double-membrane-enclosed organelle. It contains its own DNA, ribosomes, and other components required for protein synthesis, most of its proteins are encoded in the cell nucleus and imported from the cytosol There are two sub compartments in mitochondria: the internal matrix space and the intermembrane space Proteins imported into the matrix of mitochondria are usually taken up from the cytosol within seconds or minutes of their release from ribosomes. Mitochondrial proteins are first fully synthesized as precursor proteins in the cytosol & then trans located into mitochondria by a posttranslational mechanism.

96. Define Genetic engineering?

Genetic engineering is the process of manually adding new DNA to an organism. The goal is to add one or more new traits that are not already found in that organism.

97. What is Incomplete Dominance?

Incomplete dominance is a form of intermediate inheritance in which one allele for a specific trait is not completely expressed over its paired allele. This results in a third phenotype in which the expressed physical trait is a combination of the phenotypes of both alleles. Example 4 O'clock

98. Define Codominance?

A situation, when both alleles appear in the phenotype neither allele is dominant. Both alleles are expressed in heterozygous individuals. Example Blood Group

99. Define Mutation?

A mutation is a change that occurs in our DNA sequence, either due to mistakes when the DNA is copied or as the result of environmental factors such as UV light and cigarette smoke

100. Define Epistasis?

Effect of one gene dependent on the presence of one or more 'modifier genes'

101. What is Aquaporin?

Water molecules move through membranes by Osmosis (diffusion) applications

102. What are PCR applications?

1) DNA Fingerprinting 2) Genotyping 3) Genomic Cloning 4) Gene expression studies 5) Drug discovery 6) classification of organisms etc.

103. Write name of 5 antibiotics?

1) Tetra cycline 2) penicillin 3) amoxicillin 4) erythromycin 5) levofloxacin

104. Define epinephrine and how it affected by beta blocker?

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Epinephrine is a hormone that is secreted by the adrenal gland in response to physical or mental stress, from fear, and is regulated by the autonomic nervous system. The release of epinephrine causes an increase in heart rate, blood pressure, and respiratory rate. B-blockers Bind to the receptor but do not activate hormone induced effects.

105. Define codon? 2 marks

A sequence of three nucleotides which together form a unit of genetic code in a DNA or RNA molecule called codon.

106. Difference b/w klinefelter syndrome and turner syndrome. 2 marks

Klinefelter syndrome is a genetic disorder that affects males. Klinefelter syndrome occurs when a boy is born with one or more extra X chromosomes (47 XXY)

Turner syndrome is a chromosomal condition in females that lack one X chromosomes they contain 45 chromosomes.

107. How turbutalene and b blocker affect the function of epinephrine adrenaline.? 5 marks

B1- and B2-adrenergic receptors (Heart & Lungs Respectively) are coupled to G proteins (Gs), which activate adenylyl cyclase. al and a2 receptors are coupled to two other G proteins, Gq and Gi, respectively. adenylyl cyclase Inhibition & phospholipase C activation. Epinephrine bound to a2 receptors causes the arteries to constrict, cutting off circulation to the blood vessels in the intestinal tract, skin, and kidneys. (peripheral organs) Agonists (terbutaline) Mimic the function of a hormone by binding to its receptor and causing the normal response. Antagonists (B-blockers) Bind to the receptor but do not activate hormone induced effects. An antagonist acts as an inhibitor of the natural hormone (or agonist) by competing for binding sites on the receptor, thereby blocking the physiological activity of the hormone.

108. Difference b/w confluent monolyer and post confluency stages?.5 marks

A confluent monolayer is an adherent cell culture (dish, plate or flask) in which the cells have formed a single layer over the entire surface area available for growth. Once the cells have started to form clusters above the first layer or have started to lift up from the surface, the culture is past confluency stage. Cells that are repeatedly passaged at densities past confluency display decreased doubling times, decreased viabilities, & decreased ability to attach.

109. Describe the larger subunit of ribosomes? 5 marks

T (transfer) site: first site where charged tRNA lands with the help of transfer factor A (AA) site: Anticodon binds to mRNA codon, P (polypeptide) site: tRNA adds AA to polypeptide, E (exit) site: after AA transfer tRNA exits here.

1) A charged tRNA with anticodon to next codon binds A site.

2) Large subunit breaks bond between tRNA & its AA at P site.

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3) Assisted by elongation factors large subunit Catalyzes formation of peptide bond b/w the AA & the one attached to tRNA in A site (peptidyl transferase activity).

110. Define carbohydrates and discuss its classes? 3

Carbohydrates are polyhydroxy aldehyde or polyhydroxy ketones. It is further classified into 4 major groups.

The simplest monosaccharides are the two three-carbon trioses

Example : - Glyceraldehyde,an aldotriose – Dihydroxyacetone, a ketotriose

Disaccharides are those sugars which yield two molecules of the same or different monosaccharides on hydrolysis – Maltose - 2 molecules of glucose on hydrolysis

Examples. Maltose, sucrose, galactose Oligosaccharides are short chains of monosaccharide units or residues (310) joined by characteristic linkages called glycosidic linkage

Polysaccharides (Glycans) are those which yield more than 10 molecules of monosaccharides on hydrolysis Some have hundreds or thousands of units • Some polysaccharides, such as cellulose, are linear chains; others, such as glycogen, are branched

111. What are missense mutations? (2 marks)

Missense Mutations: Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein

112. What is TRANSMISSION OF SINGNAL FROM SYNAPSIS TO NEUROMUSCLE?

A neuromuscular junction (or myoneural junction) is a chemical synapse formed by the contact between a motor neuron and a muscle fiber. It is at the neuromuscular junction that a motor neuron is able to transmit a signal to the muscle fiber, causing muscle contraction.

113. What is NEGATIVE and Positive CONTROL OF TRANSCRIPTION.?

The regulatory molecule (the repressor) prevents transcription. Negative Control of Transcription:

Positive Control of Transcription: the regulatory molecule (the CRP-CAMP complex) enhances transcription.

114. What is G proteins vision 2 Activates cyclic GMP phosphodiesterase in vertebrate rod photoreceptors.

Write Role of MHC 2 the function of MHC molecules is to bind peptide fragments derived from pathogens and display them on the cell surface for recognition by the appropriate T cells.

115. Write a note on cloning 5 marks

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The term in itself means the reproduction of a biological entity which is genetically identical to the one from which it originated. The Greek term 'klon' makes us think of a sprig from a plant which when inserted into mature soil can reproduce the plant from which it was taken.

Steps of Cloning

1) The reproduction of a cell line starting from a single cell: the cells produced are histologically identical to the ones from which they originated. Cloning is also spoken of for the reproduction of DNA fragments starting from a single DNA fragment;

2) The reproduction of embryos by disaggregating or subdividing an embryo (embryo splitting) in its early stages of development when the cells are totipotent or pluripotent, i.e., capable of developing into a complete organism. The embryos obtained are then transferred in utero;

3) The reproduction of genetically identical individuals through the nucleo-transfer of a somatic cell from an embryo, fetus or an adult individual, to a denucleated oocyte. The embryos obtained are later transferred in utero. The procedure can be repeated with several nuclei taken from the somatic cells of the same individual and inserted into denucleated ova;

4) The transfer of the nucleus of an oocyte to the cytoplasm of a second denucleated oocyte, as a form of "prevention" of mitochondrial diseases. This, however, is not cloning in a strict sense. The modified oocyte is later fertilized in vitro and transferred in utero.

116. Write Types of mutations 5 marks

Somatic mutations: Occur in somatic (body) cells & passed on to the daughter cells after mitosis but not passed on to the children. Germ line mutations: Occurs in the germ line cells that give rise to gametes. A gamete with the mutation passes it on to embryo after fertilization. Point mutations: mutations of single base pairs. Chromosomal mutations: change the position or orientation of a DNA segment. Silent Mutations: Mutations result in no change in amino acids due to redundancy genetic code Nonsense mutations: Base substitution results in stop codon, (UAG) to form in the mRNA product Missense Mutations: Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein Frame-Shift Mutations: No substitutions. Single base inserted or deleted from DNA. Genetic message out of register "reading frame" changed

117. What is Nuclear pore complex 10 marks

Eukaryotic nuclear envelope is perforated by large, elaborate structures called nuclear pore complexes which selectively facilitate nuclear transport. Each pore complex contains one or more open aqueous channels through which small water-soluble molecules can passively diffuse. Each complex has a molecular mass of about 125 million & is composed of aprox 50 different proteins, called nucleoporins, arranged octagonal symmetry. Small molecules (5000 daltons or less) diffuse in so fast that the nuclear envelope can be considered to be freely permeable to them; a protein of 17,000 daltons takes 2 minutes to proteins larger than 60,000 daltons are unable to enter the nucleus. Nuclear pore contains a pathway for free diffusion 9 nm in diameter & 15 nm long, this channel occupies a small fraction of total volume of the pore complex.

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The more active the nucleus is in transcription, the greater the number of pore complexes. The nuclear envelope of a typical mammalian cell contains 3000–4000 pore complexes. Cells synthesizing DNA need to import about 106 histone molecules from the cytosol every 3 minutes to package DNA into chromatin, ie each pore complex needs to transport around 100 histone molecules/minute. Rapidly growing cell need to transport about 6 large & small ribosomal subunits/minute from nucleus, where they are produced, to the cytosol

118. What is Complement system 10 marks

Complement System. Most of the time, the binding of antibodies to antigen performs no useful function until and unless it can activate an effector mechanism. Complement system serves several effector roles. Complement system provides the actual protection from the response while the interaction of antibodies and antigen provides the specificity of the response. Put another way, antibodies "finger" the target, complement destroys it the complement system consists of some 30 proteins circulating in blood plasma. These proteins are synthesized mainly in the liver, and they account for about 5% of the globulin fraction of blood serum. Most of these are inactive until they is cleaved by a protease which, in turn, converts them into a protease. Thus many components of the system serve as the substrate of a prior component and then as an enzyme to activate a subsequent component.

The classical pathway is triggered by activation of the C1-complex (which consists of six molecule C1q and two molecules C1r and C1s), either by Ciq's binding to antibodies from classes M and G, complexed with antigens, or by its binding C1q to the surface of the pathogen. C4b, which binds covalently to sugar residues on cell-surface glycoproteins, & a smaller, inactive, fragment of C4a which diffuses away C2 is cleaved into C2b, which binds non-covalently to a site on C4b, leaving a smaller, inactive, fragment of C2a which diffuses away

This binding leads to conformational changes in C1q molecule, which leads to the activation of two cir (serine protease) molecules. Then they cleave C1s. (another serine protease) The C1-complex now binds to and splits C2 and C4, producing C2a and C4b

The complex of c4b• 2b is called "C3 convertase" because it catalyzes the cleavage of C3 C3b binds covalently to glycoproteins scattered across the cell surface. Macrophages and neutrophils have receptors for C3b and can bind the C3b-coated cell or particle.

119. Differentiate between pyrimidine and purines?

Purine is a double ring nitrogenous base while pyrimidine is a single ring nitrogenous base.

Purine contains two nitrogenous bases i.e. Adenine and guanine while pyrimidine contains three nitrogenous bases i.e. cytosine, thymine and uracil

120. Enlist requirement for PCR?

1) Target DNA 2) Two oligonucleotide primers 3) DNA polymerase, 4) dNTPs 5) PCR machine

121. DNA replication?

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In molecular biology, DNA replication is the biological process of producing two identical replicas of DNA from one original DNA molecule. This process occurs in all living organisms and is the basis for biological inheritance. DNA is made up of a double helix of two complementary strands.

122. Difference between DNA and RNA? (5)

DNA

It is double stranded molecule Cytosine pairs with Thymine It contains DeoxyRibose sugar It replicates to form new DNA It is long lived molecule

RNA

It is a single stranded molecule Cytosine pairs with Uracil It contains Ribose sugar It cannot replicates itself It has short life span except a few RNAS

123. Describe process of mitosis?

Mitosis is a type of cell division in which one parent cell divides to form two daughter cells that contain same no of chromosomes this type of cell division occurs only in somatic cells. It has phases

1) Prophase: The beginning of mitosis. so the individual chromatids become visible. Microtubules form between each pole and the chromosomes to make up a spindle.

2) Prometaphase: is marked by the disappearance of the nuclear envelope. Chromosomes begin to move toward the poles.

3) Metaphase: All the centromeres arrive at the equatorial plate. Metaphase is the best time to see the sizes and shapes of chromosomes because they are maximally condensed.

4) Anaphase: Separation of the chromatids marks the beginning of anaphase, during which the two sister chromatids move to opposite ends of the spindle. Each chromatid con-tains one double stranded DNA molecule and is now referred to as a daughter chromosome.

5) Telophase: Chromosomes stop moving at the end of anaphase, the cell enters telophase. The chromosomes begin to uncoil, the nuclear envelopes and nucleoli, reform their respective structures

124. Gel electrophoresis

A gel is made up of agarose polymer suspended in a buffer. it sits in a chamber between two electrodes, depression in the gel (wells) are filled with DNA solution, restriction enzyme 1 cuts the DNA once resulting in fragments A and B. Restriction Enzyme 2 cuts the DNA once, at a different restriction sequences. if other restriction enzymes are used , two cuts are made in the DNA. Each sample is loaded into one well in the gel. as fragments of DNA move toward the positive electrode , shorter fragments move faster than longer fragments. The fragments can be identified by gel electrophoresis on the basis of their sizes.

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Semi conservative Replication: would produce molecules with both old and new DNA but each molecule would contain one complete old strand and one new one.

Dispersive Replication: would produce two molecules with old and new DNA interspersed along each strand

125. Quaternary structure of protein

Protein quaternary structure is the number and arrangement of multiple folded protein subunits in a multi-subunit complex. It includes organizations from simple dimers to large homooligomers and complexes with defined or variable numbers of subunits.

126. Define Tautomer

Tautomers are constitutional isomers of organic compounds that readily interconvert. The concept of tautomerizations is called tautomerism. The chemical reaction interconverting the two is called tautomerization.

127. Write functions of fibriogen 2 marks

Fibrinogen, or factor I, is a blood plasma protein that's made in the liver.Fibrinogen is one of 13 coagulation factors responsible for normal blood clotting. When you start to bleed, your body initiates a process called the coagulation

128. What is Alpaktonuria?

Alkaptonuria (black urine disease, black bone disease, or alcaptonuria) is a rare inherited genetic disorder in which the body cannot process the amino acids phenylalanine and tyrosine, which occur in protein.

129. Write three stop codons?

1) UAA

2) UAG

3) UGA

130. Write three RNA Viruses?

HCV, HBV, HIV , Adeno Viruses, Influenza Virus, cytomegalo Virus etc

131. How Different types of bacteria will produce

Binary fission ("division in half") is a kind of asexual reproduction. It is the most common form of reproduction in prokaryotes especially bacteria reproduce by binary fission.

132. F. Griffith was used in his experiment? (2 marks

Griffith used two strains of Pneumococcus bacteria, type III-S (smooth) and type II-R (rough)

133. what is RFLPs and its function

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What is RFLP: RFLP is a difference in homologous DNA sequences that can be detected as fragments of different lengths after digestion of the DNA samples. Most RFLP markers are heterozygous and highly locus-specific. RFLP probe is a labeled DNA sequence that hybridizes with one or more fragments of the digested DNA sample after they were separated by gel electrophoresis

134. Difference between homozygous and heterozygous

A homozygous trait is when two of the same kind of alleles combines to form a trait. Eg. TT or tt

Heterozygous is a trait when two different kinds of alleles combine to form a trait. Eg, Tt

135. Difference between Stem cells and Specialized Cells

Stem cells differ from other kinds of cells in the body. All stem cells-regardless of their source-have three general properties: they are capable of dividing and renewing themselves for long periods; they are unspecialized; and they can give rise to specialized cell types

Groups of specialized cells cooperate to form a tissue, such as a muscle. Different tissues are in turning grouped together to form larger functional units, called organs. Each type of cell, tissue, and organ has a distinct structure and set of functions that serve the organism as a whole

136. Cam kiase Il role in movement of muscular tissue

Myosin light-chain phosphatase, more commonly called myosin phosphatase (EC 3.1.3.53), is an enzyme (specifically a serine/threonine-specific protein phosphatase) that dephosphorylates the regulatory light chain of myosin II. This dephosphorylation reaction occurs in smooth muscle tissue and initiates the relaxation process of the muscle cells. Thus, myosin phosphatase undoes the muscle contraction process initiated by myosin light-chain kinase.

The enzyme is composed of three subunits: the catalytic region (protein phosphatase 1, or PP1), the myosin binding subunit (MYPT1), and a third subunit (M20) of unknown function. The catalytic region uses two manganese ions as catalysts to dephosphorylate the light-chains on myosin, which causes a conformational change in the myosin and relaxes the muscle. The enzyme is highly conserved and is found in all organisms' smooth muscle tissue. While it is known that myosin phosphatase is regulated by rho-associated protein kinases, there is current debate about whether other molecules, such as arachidonic acid and CAMP, also regulate the enzyme.

137. What is the role of rod cell in vision?

Rod cells are photoreceptor cells in the retina of the eye that can function in less intense light than the other type of visual photoreceptor, cone cells. Rods are usually found concentrated at the outer edges of the retina and are used in peripheral vision. Activation of photopigments by light sends a signal by hyperpolarizing the rod cell, leading to the rod cell not sending its neurotransmitter, which leads to the bipolar cell then releasing its transmitter at the bipolar-ganglion synapse and exciting the synapse.

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138. Define Laminar flow cabinet principle and function?

Function: Provides sterile working area where cells can be handled without worrying about contamination.

Principal: In a laminar flow hood the air is passed through a HEPA (High Efficiency Particulates Air) filter which removes all airborne contamination to maintain sterile conditions. ... Now the sterile air flows into the working (flasking) area where you can do all your flasking work without risk of contamination.

139. Write steps involving E-coli signaling

After ligand binding change in tertiary structure of extra cellular part of Envz leads to structural change in its cytoplasmic domain making it a kinase (auto..). EnvZ-P can now phosphor-ilate OmpR (responder) outside signal in and amplified. Receptor conformational change after ligand binding which activates kinase activity. Phosphorilation alters responder function. Signal amplified. Transcription factor activated. Protein synthesis results in altered cell activity.

Ligand binding causes a structural change permitting G protein to bind receptor. Binding of G protein to activated receptor causes it to exchange GDP for GTP (ligand releases).

140. How antibody help the surface to produce B cells

Each B cell lymphocyte produces one unique antibody against one unique epitope. Once antibodies with sufficient specificity to the epitope can be encoded, the B cell begins to release antibodies into the bloodstream. These antibodies then bind specifically with the foreign molecule and allow the immune system to eliminate the molecule from the system.

141. Define Gene linkage? (2)

Linkage is defined genetically as the failure of two genes to assort independently. Linkage occurs when two genes are close to each other on the same chromosome

142. VNTRs stand for?

Variable Number Tandem Repeats

143. What are the steps of initiation of translation process?

a)Shine Dalgarno sequence:

In prokaryotes the rRNA of the small ribosomal subunit first binds to a complementary ribosome binding site upstream start codon AGGAGG this is called shine Dalgarno sequence on the mRNA.

b) Anticodon of methioninie charged to tRNA and bindes to start the codon (Methionine can be removed after translation).

c) In the small subunit rRNA stabilizes H-bonding between tand m RNA.

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d) Large subunit facilitated by initiation factors binds the complex with methhionine charged tRNA at the P site.

144. What are elongation steps in translation process?

a) A charged tRNA with anticodon to next codon binds A site.

b) Large subunit breaks bond between tRNA & its AA at P site.

c) Assisted by elongation factors large subunit Catalyzes formation of peptide bond between the AA and the one attached to tRNA in A site (peptidyl transferase activity).

145. What are termination steps in translation process?

a)When stop codon enters A site elongation is terminates.

b) Stop codons bind release factor, which hydrolysis bond between polypeptide chain and tRNA at P site.

c) Protein folding information is a function of AA sequence.

146. Define Somatic mutations?

It Occur in somatic (body) cells & passed on to the daughter cells after mitosis but not passed on to the children.

147. Define Germ line mutations?

It Occurs the germ line cells that give rise to gametes. A gamete with the mutation passes it on to embryo after fertilization.

148. What are Point mutations?

These are Mutations of single base pairs.

149. Write function of chromosomal mutations?

It changes the position or orientation of a DNA segment.

150. Define silent mutation?

The Mutations which results in no change in amino acids due to redundancy genetic code.

151. Define Missense Mutations with example?

These are Base substitution mutations that change the genetic message i.e. one amino acid substitutes for another in the protein. For example Sickle-cell disease results from a defect in human B-globin gene, a protein in human red blood cells that carries oxygen.

152. Define Frame-Shift Mutations?

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In which No substitutions accurs a Single base inserted or deleted from DNA. Genetic message out of register "reading frame" changed.

153. What is epigenetics?

Epigenetics is the study of heritable changes in gene expression that do not involve changes in the DNA sequence.

154. Differentiate between Positive Control of Transcription and negative Control of Transcription?

Positive vontrol of Transcription is the trascription in which regulatory molecule (the CRP-CAMP complex) enhances transcription. While negative Control of Transcription in which the regulatory molecule (the repressor) prevents transcription.

Write Three Systems of regulations of operon?

• An inducible operon regulated by a repressor protein

• A repressible operon regulated by a repressor protein

• An operon regulated by an activator protein

155. Write two essential sequence of the promoter?

Promoter has two essential sequences. 1) Recognition sequence (RNA polymerase). 2) TATA box where DNA begins to denature.

156. How pre-mRNA (freshly transcribed) is processed to make it functional?

1-Introns are removed during RNA maturation. 2) The transcribed pre-mRNA is altered by the addition of a G cap (modified GTP) at the 5' end to protect RNA and facilitate translation. 3) After the last codon a poly A tail (100-300) is added at the 3' end for export and protection.

157. What is magnetism?

Magnetism can be a Signal which in some bacteria and birds orient themselves to Earth's magnetic poles.

158. Define Signal Transduction?

The entire signaling process, from signal detection to final response is called signal transduction pathway/ signaling Cascades

159. What are types of protein in signaling pathway?

1) Relay proteins: pass the message to the next signaling component in the chain.

2) Messenger proteins: carry the signal from one part of the cell to another (cytosol to nucleus).

3) Adaptor proteins: link one signaling protein to another, without themselves conveying a signal.

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4) Amplifier protein: either enzymes or ion channels increase the signal they receive, (Produce large amounts of small intracellular mediators/activate signaling proteins (adenylyl cyclase).

5) Transducer proteins: convert the signal into a different form (adenylyl cyclase). Gnaling.

6) Bifurcation proteins: spread signal from one pathway to another.

7) Integrator proteins: receive signals from 2++ pathways integrate them before relaying a signal onward.

8) Latent gene regulatory proteins: activated at the cell surface then migrate to nucleus to stimulate gene transcription.

160. Write at least one function of scaffolds protein?

Many cells use scaffold proteins to enhance speed efficiency & specificity of response.

161. What is abbreviation of GPK, GP, PP, GS?

Glycogen phosphorylase kinase (GPK), glycogen phosphorylase (GP) phosphoprotein phosphatase (PP), glycogen synthase (GS)

162. What is difficult between Agonists and antagonists?

Agonist (terbutaline) Mimic the function of a hormone by binding to its receptor and causing the normal response while Antagonist (B-blockers) Bind to the receptor but do not activate hormone induced effects.

163. What are the causes of CAMP?

CAMP rise causes membrane proteins to permit a massive flow of water from the blood into the intestinal lumen, causing diarrhea (water loss) resulting in death from dehydration.

164. How CML formed?

Chronic myelogenous leukemia (CML) results from a chromosomal translocation between chromosome 9 and 22

165. What Is Bruton's tyrosine kinase,?

Bruton's tyrosine kinase, a protein whose mutation is associated with X-linked a gammaglobulinemia.

166. What is the function of Erythropoietin?

Erythropoietin stimulates erythrocyte precursor cells to survive, proliferate, and differentiate, is associated with Jak2

167. Write function of prolactin?

Prolactin stimulates breast cells to produce milk Activated STAT5 stimulates the transcription of genes that encode milk proteins.

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168. How many receptors are required for signaling mentioned their name?

Two types of receptors are required for signaling Both type-l and type-ll receptors, required for signaling.

169. What is abbreviation of PDGF and EGF?

platelet-derived growth factor (PDGF) and epidermal growth factor (EGF).

170. Write characteristics of Jak stat signaling?

Jak-STAT Signaling.Increase antigen presentation of macrophages.Activate & increase lysosome activity macrophages. Cells to increase expression of MHC class .Promotes adhesion and binding required for leukocyte migration. Promotes NK cell activity. Activate inducible Nitric Oxide Synthase.Jak-STAT one of the most direct signaling. Ligand binding induces receptor to dimerize. Binding brings Jaks close to crossphosphorylate each other, increasing kinase activity. The Jaks then phosphorylate receptors, creating STAT docking sites

171. Define antigens?

Substances that activate adaptive immune specific (AIS) are called antigens.

Write names of five class of antibodies?

5 classes of antibodies, IgA, IgD, IgE, IgG, & IgM,

172. Explain immunological memory?

Immunological Memory: The adaptive immune system, like the nervous system, can remember prior experiences. First exposure to an antigen A: Immune response appears after several days, rises rapidly and exponentially, and then, more gradually, declines. Secondary immune response: Lag period is shorter, and the response is greater indicating animal remembers its first exposure to antigen A. If animal is given a different antigen B instead of second injection of antigen A the response is typical primary not secondary, immune response.

173. What are Cellular Basis of Immunological Memory?

When naïve lymphocytes are stimulated by their specific antigen, they proliferate and differentiate. Most become effector cells which function & then die, while others become long-lived memory cells.

174. What is mechanism of class switching?

Class Switching: B-Cell initially produces IgM (antibody's heavy chain is encoded by the first constant region gene, the u segment) molecules, which recognise specific antigenic determinant. When B cell becomes a plasma cell another deletion occurs in the cell's DNA, which position heavy-chain variable region gene next to a different constant region (y, ε, or A)

175. What are the types of MHC molecules?

There are two types of MHC molecules MHC class I and MHC class II with distinct functions.

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176. What is genetic diversity of MHC classes?

3 genes for class I proteins (HLA-A, B &C). 3 genes for class II proteins (HLA-DP, DQ &DR)6 genes each for both the classes.

177. What happened after T cell activation?

Once a helper T cell is activated by an antigen-presenting cell to become an effector cell, it then help activate other cells.

178. What is TH1 and TH2 explain them?

When activated by an antigen-presenting cell, a naïve helper T cell can differentiate into either of two distinct types of effector helper cell, called TH1 and TH2. TH1 cells mainly help activate macrophages and cytotoxic T cells, whereas TH2 cells mainly help activate B cells.

179. Write classes of proto-oncogene?

Classes: Growth Promoting Oncogenes, Growth-Inhibiting Cancer Suppressor Genes, DNA Repairing Genes, Apoptosis Genes.

180. LDL particles (3 marks)

Receptor-mediated endocytosis is the method by which cholesterol is taken up by most mammalian cells. Water-insoluble cholesterol and triglycerides are packaged by liver cells into lipoprotein particles, which are then secreted into the bloodstream to provide body tissues with lipids. One type of these particles, called low-density lipoproteins, or LDLs, must be taken up by the liver for recycling. This uptake also occurs via receptor-mediated endocytosis.

This process begins with the binding of LDLs to specific receptor proteins on the cell surface. Once engulfed by endocytosis, the LDL particle is freed from the receptors. The receptors segregate to a region of the vesicle that buds off to form a new vesicle, which is recycled to the plasma membrane. The freed LDL particle remains in the original vesicle, which fuses with a lysosome in which the LDL is digested and the cholesterol made available for cell use. Persons with the inherited disease familial hypercholesterolemia (-emia, "blood") have dangerously high levels of cholesterol in their blood because of a deficient receptor for LDL.

181. Gene regulatory expression or housekeeping cells (3 marks)

There are two alternatives-transcriptional regulation and posttranscriptional regulation can be distinguished by examining the actual RNA sequences made within the nucleus of each cell type. Such analyses indicate that for some proteins, the mechanism of regulation is differential gene transcription.

Both brain and liver cells, for example, transcribe "housekeeping" genes-those that encode proteins involved in the basic metabolic processes that occur in every living cell, such as glycolysis enzymes. But liver cells transcribe some genes for liver-specific proteins, and brain cells transcribe some genes for brain-specific proteins. And neither cell type transcribes the

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genes for proteins that are characteristic of muscle, blood, bone, or the other specialized cell types in the body.

182. Simple staining and gram staining (2 marks)

In simple staining, the bacterial smear is stained with a single reagent, which produces a distinctive contrast between the organism and its background. Basic stains with a positively charged chromogen are preferred because bacterial.

Gram stain or Gram staining, also called Gram's method, is a method of staining used to distinguish and classify bacterial species into two large groups: gram-positive bacteria and gram-negative bacteria.

183. Signaling sequence (2 marks)

Signal sequences are N-terminal extensions of newly synthesized secretory and membrane proteins. In eukaryotes, signal sequences direct the insertion of proteins into the membrane of the endoplasmic reticulum and are usually cleaved off by signal peptidase.

184. Cytokinesis (2 marks)

Cell separation, or cytokinesis, begins 20 minutes after chromosome replication is finished. The first event of cytokinesis is a pinching in of the plasma membrane to form a ring similar

185. Differentiate between Euchromatin and Heterochromatin (3 marks)

Under a microscope, two kinds of chromatin can be distinguished in the stained interphase nucleus: Euchromatin and heterochromatin. Euchromatin is diffuse and stains lightly; it contains the DNA that is transcribed into mRNA. Heterochromatin stains densely and is generally not transcribed; any genes that it contains are thus inactivated.

186. How DNA is packaged in Cell. (2 marks)

DNA is tightly packed up to fit in the nucleus of every cell. As shown in the animation, a DNA molecule wraps around histone proteins to form tight loops called nucleosomes. These nucleosomes coil and stack together to form fibers called chromatin.

187. What is primary culture? (2 marks)

Primary culture refers to the stage of the culture after the cells are isolated from the tissue and proliferated under the appropriate conditions until they occupy all of the available substrate (i.e., reach confluence). ... After the first subculture, the primary culture becomes known as a cell line or subclone.

188. Why the cell is important? (3 marks)

Cells are the basic building blocks of all living things. The human body is composed of trillions of cells. They provide structure for the body, take in nutrients from food, convert those nutrients into energy, and carry out specialized functions.

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189. Name three RNA viruses (3 marks)

Human diseases causing RNA viruses include:

Orthomyxoviruses, Hepatitis C Virus (HCV), Ebola disease, SARS, influenza, polio measles and retrovirus including adult Human T-cell lymphotropic virus type 1 (HTLV-1) and human immunodeficiency virus (HIV).

190. Which antibody is present on the surface of b cells? (2 marks)

IgM is not only the first class of antibody to appear on the surface of a developing B cell. It is also the major class secreted into the blood in the early stages of a primary antibody response, on first exposure to an antigen.

191. what is palindromic sequence (3 marks)

A palindromic sequence is a nucleic acid sequence in a double-stranded DNA or RNA molecule wherein reading in a certain direction on one strand matches the sequence reading in the opposite direction on the complementary strand.

192. Difference between Homotypic and Heterotypic. (3marks)

In most cases, the binding of cells in a tissue is Homotypic; that is, the same molecule sticks out of both cells, and the exposed surfaces bind to each other. But heterotypic binding (of cells with different proteins) also can occur. For example, when the mammalian sperm meets the egg, different proteins on the two types of cells have complementary binding surfaces.

193. What was the technique used by Mendel in his experiment (2)

Emasculation and bagging were the techniques used by Mendel to perform hybridisation. If the female flower is bisexual, removal of anthers from the flower bud before the anther, dehisce using a pair of forceps. This is called as emasculation.

194. Active and passive transport (3 marks)

Active transport is the movement of molecules or ions against a concentration gradient (from an area of lower to higher concentration), which does not ordinarily occur, so enzymes and energy are required. Passive transport is the movement of molecules or ions from an area of higher to lower concentration.

195. What Is CANNA PK 13/14? (3 marks)

CANNA PK 13/14 is a flowering stimulator high in phosphorus and potassium manufactured by CANNA. PK 13/14 ensures stronger cell formation, the production of enough sugars in the flowering stage, and increases plants osmotic value which improves the volume, weight and taste of your crop.

196. Difference b/w saturated and unsaturated fatty acids

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Saturated fatty acids lack double bonds between the individual carbon atoms, while in unsaturated fatty acids there is at least one double bond in the fatty acid chain. Saturated fats tend to be solid at room temperature and from animal sources, while unsaturated fats are usually liquid and from plant sources.

197. Role of DNA in living organism

DNA is vital for all living beings - even plants. It is important for inheritance, coding for proteins and the genetic instruction guide for life and its processes. DNA holds the instructions for an organism's or each cell's development and reproduction and ultimately death.

198. Role of protein in living organism

Protein is a nutrient your body needs to grow, as well as to support and maintain your life. ... For example, proteins help cells develop and communicate, act as enzymes and hormones, conduct the transport of nutrients throughout your bloodstream and repair damaged tissue. In other words, you can't live without protein.

199. Process of developing of T-cell

T cells are derived from haematopoietic stem cells that are found in the bone marrow. The progenitors of these cells migrate to and colonise the thymus. ... The majority of cells in the thymus give rise to aß T cells, however approximately 5% bear the yo T cell receptor (TCR).

200. Cam kinase II function in cellular movement

CaM kinase Il regulates neural excitability, phosphorylating Synapsin I (see Drosophila Synapsin), a protein which functions in regulating vesicular movement. Vesicles release neurotransmitters at the synapse, and their dynamics is the focus of regulation of neural excitability.

201. Function of Adaptive Immune system.

The adaptive immune system, also known as the specific immune system, is composed of highly-specialized systemic cells and processes that eliminate or prevent pathogenic growth. The adaptive immune system works to protect and heal the body when the innate immune system fails

202. Clinical use of cell culture

Cell culture includes processes for isolating cells from tissue and maintaining, proliferating, and/or differentiating the cells in a culture vessel containing medium. ... Cultured cells are not only used for basic research in biology and molecular biology, but also for drug discovery and regenerative medicine.

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